

**Time:** 5 minutes**Outcome:** Use Short-Run Firm Choice, Profit/Loss & Shutdown.**Difficulty:** ●●○ Intermediate

Short-Run Firm Choice, Profit/Loss & Shutdown



THE CORE IDEA

A firm with a differentiated product faces downward-sloping demand. At an interior optimum, choose quantity where marginal revenue (MR) crosses marginal cost (MC) from above, then read price from demand. Compare price with average total cost (ATC) for profit or loss and with average variable cost (AVC) for the short-run shutdown decision.



HOW TO RECOGNIZE IT

- Choose the operating output where MR crosses MC, then read price from demand.
- Compare price with ATC to identify profit or loss.
- Compare revenue with variable cost before deciding whether to shut down.

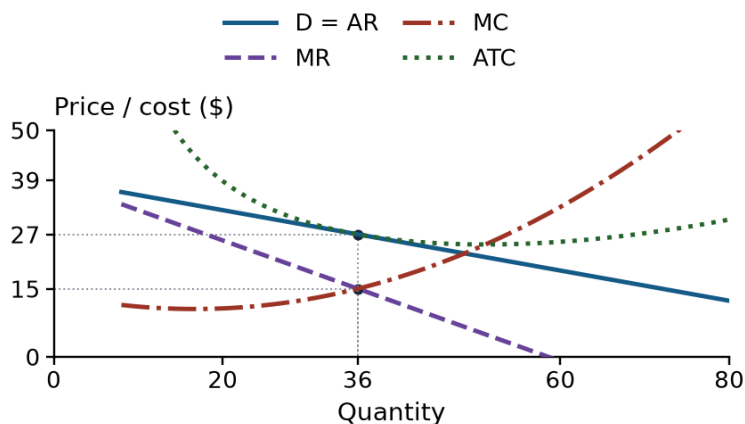


WATCH OUT

Separate profit from shutdown. A fixed-cost increase can create an economic loss without changing marginal cost or the benefit of operating; compare revenue with variable cost.



WORKED EXAMPLE: SHORT-RUN FIRM DECISION



The displayed baseline has $Q = 36$ and $P = ATC = \$27$, so economic profit is zero. Consider a higher fixed lease cost of \$108, with demand and variable costs unchanged. Output and price stay at that baseline, but ATC rises by \$3 and the loss is \$108. The firm still operates if revenue covers variable cost; loss alone does not imply shutdown.



CHECK YOURSELF

A firm earns a loss because P is below ATC . What additional cost comparison is needed to decide whether it should shut down in the short run?

**READY?**

Return to the game and master this concept.